

Wei Lu

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EDUCATION

Rotman School of Management, University of Toronto , Toronto, Canada <i>Ph.D. in Quantitative Marketing</i> <i>Committee: Avi Goldfarb (Chair), Daniel Goetz, Nitin Mehta, Matthew Osborne</i>	Expected 2023
Toulouse School of Economics , Toulouse, France <i>M.A. in Economics</i>	2017
Central University of Finance and Economics , Beijing, China Victoria University , Melbourne, Australia <i>B.A. in Economics and B.Bus., joint degree (with distinction)</i>	2015
University of California, Los Angeles , Los Angeles, California, USA Exchange student	2014

RESEARCH INTEREST

Topic: Economics of Digitization, Platform Design, Empirical Industrial Organization
Method: Structural Modeling, Causal Inference, Machine Learning, Natural Language Processing

WORKING PAPERS

Product Development and Platform Design (Job Market Paper)

Wei Lu, Avi Goldfarb, and Nitin Mehta

*** Winner, ISMS Doctoral Dissertation Proposal Competition, 2021 ***

Peer Effects from Friends and Strangers: Evidence from Random Matchmaking in an Online Game (with Daniel Goetz) [\[SSRN\]](#)

Accepted at EC'22: Proceedings of the 2022 ACM Conference on Economics and Computation

Recipient of TD MDAL Research Grant, 2021

Manuscript finished and ready to be submitted to Management Science

Increasing Dynamic App Engagement with Peak-End Effects (with Daniel Goetz)

Recipient of BEAR/BI-Org Ph.D. Research Awards, 2021

Evaluation Inflation: Consumers Give Higher Product Ratings on Transaction Sites than Non-transaction Sites (with Claire Tsai and Ying Zeng)

Recipient of TD MDAL Research Grant, 2021

Manuscript finished and ready to be submitted to Journal of Marketing Research

WORK IN PROGRESS

Food Purchasing Patterns and Nutrition Quality in the Time of Pandemic: Evidence from COVID-19 and Lockdowns

Wei Lu, Guanlan Hu, David Soberman, and Mary L'Abbé

Motivated Reasoning in the Field: Polarization in Precedent, Prose, and Policy in U.S. Circuit Courts

Wei Lu, Daniel L. Chen

CONFERENCE PRESENTATIONS

The 23rd ACM Conference on Economics and Computation (EC'22)*, Boulder, CO	2022
ISMS Marketing Science Conference, Chicago, IL	2022
TD MDAL Research Roundtable, Toronto, ON	2022
ISMS Marketing Science Conference, Rochester, NY	2021
Trans-Atlantic Doctoral Conference, London, UK	2021
EMAC Conference, Madrid, Spain* (* presented by coauthor)	2021
The 2018 Meetings of The Canadian Law and Economics Association, Toronto, ON	2018

GRANTS, HONORS & AWARDS

Winner, ISMS Doctoral Dissertation Proposal Competition	2021
BEAR/BI-Org Ph.D. Research Awards (CAD \$6900)	2021
TD MDAL Research Grant (CAD \$5000, with Daniel Goetz)	2021
TD MDAL Research Grant (CAD \$5000, with Claire Tsai, and Ying Zeng)	2021
NBER Digitization Tutorial Fellow	2021
ISMS Marketing Science Conference Doctoral Consortium Fellow	2020
NBER Young Scholars Workshop on the Economics of Artificial Intelligence Fellow	2019, 2020
Quantitative Marketing and Structural Econometrics Workshop Participant	2019
University of Toronto Fellowships	2017-2022
Director's Scholarship	2017-2019
Best Undergraduate Thesis Award	2015
Outstanding Graduate of Beijing City	2015

TEACHING EXPERIENCE

Project Coach, University of Toronto

Management Analytics Practicum (<i>Master of Management Analytics</i>), Instructor: Dmitry Krass	2021
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Teaching Assistant, University of Toronto

Pricing (<i>Undergraduate</i>), Instructor: Matthew Osborne	2020
Statistics for Management (<i>MBA</i>), Instructor: Sridhar Moorthy	2020
Introduction to Big Data Analysis (<i>MMA</i>), Instructor: Daniel Goetz	2019-2021
Analytics for Marketing Strategy (<i>MMA</i>), Instructor: Nitin Mehta	2019
Principles of Marketing (<i>Undergraduate</i>), Instructor: Zachary Zhong	2019
Consumer Behaviour (<i>Undergraduate</i>), Instructor: Scott Hawkins	2019

SELECTED GRADUATE COURSEWORK

Economics

Competition and Market Strategies (TSE)
Microeconomics I (TSE)
Microeconomics II (TSE)
Microeconomics II (UofT)
Econometrics I (UofT)
Econometrics I (TSE)
Econometrics II (UofT)
Industrial Organization II (UofT)
The Econometrics of Program Evaluation (TSE)
Empirical Applications of Economic Theory (UofT)
Methods for Empirical Microeconomics (UofT)
Statistical Methods for Machine Learning II (UofT, audited)

Instructors

Doh-Shin Jeon
Renato Gomes
Patrick Rey
Martin Osborne
Jiaying Gu and Yuanyuan Wan
Eric Gautier
Adonis Yatchew & Victor Aguirregabiria
Victor Aguirregabiria
Sylvain Chabé-Ferret
Eduardo Souza-Rodrigues
Arthur Blouin
David Duvenaud & Jesse Bettencourt

Marketing

Marketing Strategy I (UofT)
Marketing Strategy II (UofT)
Advanced Topics in Marketing Strategy (UofT)
Judgement and Decision Making Theory (UofT)
Experimental Economics (UofT)
Empirical Methods in Marketing (UofT)

Pankaj Aggarwal & Claire Tsai
Sridhar Moorthy & Avi Goldfarb
David Soberman
Dilip Soman
Tanjim Hossain
Matthew Osborne

MISCELLANEOUS

Computer Skills: Python, R, MATLAB, Stata, SPSS, SAS.

Languages: English (fluent), Mandarin (native), French (beginner)

REFERENCES

Avi Goldfarb (Supervisor)

Rotman Chair in Artificial Intelligence and Healthcare
Professor of Marketing
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ABSTRACTS

Product Development and Platform Design (Job Market Paper)

Wei Lu, Avi Goldfarb, and Nitin Mehta

Recipient of ISMS Doctoral Dissertation Award 2021

Abstract: We study how platform-wide pricing policy changes can incentivize firm-level innovations. Using a unique dataset of beta apps on a video game marketplace, Steam, we study how commission schemes can change updates and release decisions of developers. We use transformer-based neural network language models and supervised machine learning algorithms to classify updates into Feature, Regular, and Bug fix updates. We document positive correlations between past product updates and consumer demand, current updates, release decisions, and past reputation and current firm decisions. We find substantial costs for producing updates and releasing the product by constructing and estimating a structural model. We examine the impact of varying platform commission rates on product development decisions and find that lowering commissions lead to more updates but fewer releases.

Peer Effects from Friends and Strangers: Evidence from Random Matchmaking in an Online Game (with Daniel Goetz)

Extended Abstract at EC'22: Proceedings of the 2022 ACM Conference on Economics and Computation

Recipient of TD MDAL Research Grant, 2021

Manuscript finished and ready to be submitted to Marketing Science

Abstract: The product adoption decisions of friends and strangers can both strongly influence consumers' purchase behaviour, but their importance relative to each other is unknown. In this paper, we use data on microtransactions in an online gaming app to show that the marginal effect of friends' product adoption is roughly twice as large as the marginal effect of strangers' product adoptions, but that the two types of peer effects are substitutes. Our empirical strategy leverages the game's quasi-random matchmaking system to identify the causal effect of observed adoptions on in-game purchasing. An estimated network diffusion model implies that peer effects from strangers alone can account for 69% of the total lift in product adoptions from social interactions. Our results have managerial implications for optimal product seeding, and we use our network model to evaluate alternative seeding strategies that leverage peer effects from friends versus strangers. Finally, we provide evidence that both types of peer effect increase product awareness, but that peer effects from friends also generate a mere exposure effect.

Increasing Dynamic App Engagement with Peak-End Effects (with Daniel Goetz)

Recipient of BEAR/BI-Org Ph.D. Research Awards, 2021

Abstract: The peak-end effect in consumer psychology predicts that individuals will re-engage with an activity sooner if the activity ends with a relatively positive experience. Using millions of unique usage sessions from two online apps, we document that an exogenously allocated positive experience makes consumers more likely to end a usage session with the app, even as it shortens the duration until they begin their next usage session. Our results show that leveraging the peak-end effect leads to shorter but more frequent usage sessions, creating a tradeoff in-app engagement.

Evaluation Inflation: Consumers Give Higher Product Ratings on Transaction Sites than Non-transaction Sites (with Claire Tsai and Ying Zeng)

Recipient of TD MDAL Research Grant, 2021

Manuscript finished and ready to be submitted to Journal of Marketing Research

Abstract: Prior research in online reviews has focused on describing their distributions (e.g., J-shape, positive skewness). The present research adds to this literature by documenting how posting on transaction sites systematically impacts consumer reviews and providing causal evidence for the underlying mechanism. Specifically, in a series of five experiments with randomization ($N = 1,518$) and archival data analysis across eight platforms ($N = 275,851$ products based on 500 million reviews), we found that consumers give higher ratings and more perfect scores on transaction sites (e.g., Amazon, Doordash) than they otherwise would on non-transaction sites (e.g., Goodreads, Yelp) or in private, while randomizing or controlling for products. Our experimental design ruled out factors that prior literature suggested could influence online review distributions, including selection biases, frauds, financial incentives, etc. Our experiments demonstrated that the effect was driven by consumers perceiving reviews on transaction sites as direct feedback to the sellers. This perception makes consumers more sensitive to sellers' desires for perfect score and conform to this perceived norm by giving higher ratings. Our findings contribute to the emerging consumer psychology literature in review creation and social influence in a digitized marketplace.